

Soroush Vahidi
Department of Computer Science
Ying Wu College of Computing
New Jersey Institute of Technology
Newark, NJ 07102, USA

September 28, 2026

Editors of *Performance Evaluation*

Dear Editors of *Performance Evaluation*,

I am pleased to submit the manuscript “How Portable Are LLM-Serving Scheduler Rankings Across Workloads, Operating Regions, and Metrics?” for consideration as an original research article in *Performance Evaluation*.

The manuscript studies whether comparative rankings of LLM-serving schedulers remain portable when the workload source, operating region, or evaluation metric changes. Its main methodological contribution is to treat comparative scheduler-ranking portability itself as an object of performance evaluation, rather than to propose another scheduling policy. The study uses three independently released workload sources, six calibrated operating regions, and 9,360 unique configurations, each executed twice.

The results are intentionally balanced. Cross-source rankings are stable for some source pairs, but 36/990 pairwise comparisons show statistically supported practical reversals, all concentrated in one policy pair with BurstGPT on one side. Cross-metric portability is substantially lower: median Kendall’s tau-b is 0.419 and top-policy agreement is 31.9%. A selected physical validation did not reproduce the largest simulated reversal, while the stable-control comparison preserved its qualitative ordering.

The manuscript fits *Performance Evaluation*’s scope through its focus on performance evaluation of computing systems, scheduling and resource allocation, trace-driven simulation, workload characterization, statistical analysis, and AI-based services. Relative to workload-sensitive dispatching studies, including the data-center dispatching analysis of Yildiz et al. (2026), LSSP evaluates the portability of an entire comparative LLM-serving scheduler ranking across independent workload sources, operating regions, and metrics, with explicit ranking uncertainty, practical reversal detection, benchmark sample-complexity analysis, and selected physical validation.

Artifacts are publicly available through GitHub, Hugging Face, and Zenodo:

<https://github.com/SoroushVahidi/llm-serving-scheduler-robustness-benchmark>
<https://huggingface.co/datasets/SoroushVahidi/llm-serving-scheduler-portability>
<https://doi.org/10.5281/zenodo.22306798>

As described in the manuscript, the release includes source code, provenance, selected canonical result tables, and the reduced RQ6 analysis result; it does not claim to publish all raw per-execution records.

This manuscript is original, is not under consideration elsewhere, and I approve its submission as the sole author. Relevant competing-interest information and other required declarations are included in the manuscript.

Sincerely,

Soroush Vahidi
Ph.D. Candidate in Computer Science
Department of Computer Science
Ying Wu College of Computing
New Jersey Institute of Technology
sv96@njit.edu